7. If $\rho(A)=r$ then which of the following is correct?
   (a) all the minors of order r which does not vanish
   (b) A has at least one minor of order r which does not vanish
   (c) A has at least one $(r+1)$ order minor which vanishes
   (d) all $(r+1)$ and higher order minors should not vanish

8. If $A=\begin{pmatrix}1\\2\\3\end{pmatrix}$ then the rank of AA^T is
   (a) 0 (b) 1
   (c) 2 (d) 3

9. If the rank of the matrix $\begin{pmatrix}\lambda & -1 & 0\\ 0 & \lambda & -1\\ -1 & 0 & \lambda\end{pmatrix}$ is 2, then λ is
   (a) 1 (b) 2
   (c) 3 (d) only real number

10. The rank of the diagonal matrix $\begin{pmatrix}1 & & & & \\ & 2 & & & \\ & & -3 & & \\ & & & 0 & \\ & & & & 0\end{pmatrix}$
   (a) 0 (b) 2
   (c) 3 (d) 5

11. If $T=\begin{array}{c}\\A\\B\end{array}\begin{array}{c}\begin{matrix}A & B\end{matrix}\\\begin{pmatrix}0.7 & 0.3\\ 0.6 & x\end{pmatrix}\end{array}$ is a transition probability matrix, then the value of x is
   (a) 0.2 (b) 0.3
   (c) 0.4 (d) 0.7

12. Which of the following is not an elementary transformation?
   (a) $R_i \leftrightarrow R_j$ (b) $R_i \rightarrow 2R_i+2C_j$
   (c) $R_i \rightarrow 2R_i-4R_j$ (d) $C_i \rightarrow C_i+5C_j$

13. If $\rho(A)=\rho(A,B)$ then the system is
   (a) Consistent and has infinitely many solutions
   (b) Consistent and has a unique solution
   (c) Consistent
   (d) inconsistent

14. If $\rho(A)=\rho(A,B)=$ the number of unknowns, then the system is
   (a) Consistent and has infinitely many solutions
   (b) Consistent and has a unique solution
   (c) inconsistent
   (d) consistent

15. If $\rho(A)\neq\rho(A,B)$, then the system is
   (a) Consistent and has infinitely many solutions
   (b) Consistent and has a unique solution
   (c) inconsistent
   (d) consistent

16. In a transition probability matrix, all the entries are greater than or equal to
   (a) 2 (b) 1
   (c) 0 (d) 3

17. If the number of variables in a non-homogeneous system $AX=B$ is n, then the system possesses a unique solution only when
   (a) $\rho(A)=\rho(A,B)>n$
   (b) $\rho(A)=\rho(A,B)=n$
   (c) $\rho(A)=\rho(A,B)<n$
   (d) none of these

18. The system of equations $4x+6y=5,\ 6x+9y=7$ has
   (a) a unique solution
   (b) no solution
   (c) infinitely many solutions
   (d) none of these

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